Module: Java Web Development Integration Project

Curriculum	Computer science	
Code Module	ECUEF-INF1-231	
Semester	2	
Tongue	French	
ECTS	7	
Load	ET: 42 h	EST: 140 h

Module Manager	Abdelkarim Mars
Teaching unit	Integration and development project
Prerequisite	Java, php framework, Software and database engineering
Target Audience	Level 1

Objectives of the teaching module

At the end of this module, the learner will be able to work in a team to develop a cross-platform application by applying the basic concepts of the Java and PHP Framework development modules and the best practices of software engineering.

Organization and Evaluation Method:

Individual grade=40% (average of the Java phase tracking grades) + 40% (average of the web phase tracking grades) + 10% GL course sprint1 grade + 10% GL course sprint2 grade

This module consists of 42 hours divided into several Sprints

Sprint 0: Project study

Sprint 1: Development of part of the application with web2.0 technologies. Sprint 2:

Development of part of the application with technology

Java

Method of evaluation:

PIDEV average = 30% team score + 70% individual score

Team score =20%GL score+ 20%Engineering skills validation score+ 20%Integration score between team member / java sprint+20%Integration score between team member/ web sprint + 20%Final integration score (java, web)

Learning outcomes:

At the end of this module, the student should be able to:

	Learning outcomes	**Level of in-depth study (*)**
AA1	Apply the agile Scrum method in an IT project.	3
AA2	Use a collaborative work platform.	3
AA3	Manipulate the graphical part of a cross-platform application.	3
AA4	Present the work developed as a team in a relevant and convincing way.	4
AA5	Create basic functionality of a cross-platform application.	5
AA6	Create advanced features of a cross-platform application.	5
AA7	Implement a multi-platform solution that leverages the same database.	5
AA8	Defend the application	6

(*) : 1-Knowledge | 2-Understanding | 3-Application | 4-Analysis | 5-Synthesis | 6-Evaluation

Detailed content:

CONTENT	LEARNING SITUATIONS	RUNTIME
Sprint0: Project Study: • Choose the theme • Choose the topic • Compose work teams • Identify the different modules of the project • Specify an integrated project through the Product Backlog. • Analyze an integrated project via analysis diagrams • Plan the sprints of an integrated project • Developing the overall architecture of a project	**Coaching**	**6 hours**
Sprint1: Development of part of the application with web2.0 technologies: • Plan and organize the activity of • development of an increment of the integration project through the Backlog sprint • Design Project Increment 1 • Estimate and evaluate the workload of Sprint 1 through Scrum artifacts • Validate increment 1 with the story tests. • Develop the basic functionalities of the different entities. • Apply input controls on entered data • Apply singleton design patterns • Create graphical interfaces • Create the advanced feature part	**Project**	**15 hours**

Sprint 2: Development of part of the application with Java technology: • Plan and organize the development activity of an increment of the integration project through the Backlog sprint • Design Project Increment 2 • Estimate and evaluate the workload of Sprint 2 through Scrum artifacts • Apply design patterns • Develop the basic functionalities of the different entities • Apply input controls on entered data • Create graphical interfaces • Create the advanced feature part (external bundles, APIs and business part)	**Project**	**15 hours**
Validation	**Project demonstration** **Commercial Defense**	**6 hours**

Assessment methods

The student is evaluated according to the table below. All evaluations will be according to a criterion-based evaluation grid

LEARNING OUTCOMES	EVALUATION METHODOLOGY	SESSION
Relate the learning outcomes of the different modules (Java, php framework, Software Engineering, database)	Final Validation	-throughout the project
Apply the agile Scrum method in integration projects.	Validation 2 and Validation 3	GL sessions
Create advanced features of a cross-platform application.	Validation (Trail 3 + Trail 5) sprint Web/Java Phase	Session 6 + Session 9

Implement a multi-platform solution that leverages the same database.	Validation 3 and Validation 4	Session 6-7-8-9-10-11-12
Defend the choice of subject.	Final validation	Session 14
Use a collaborative work platform.	Each follow-up session + validation 1 + validation 2	Throughout the project
Manipulate the graphical part of a cross-platform application.	Follow-up 4+follow-up 8+Validation 1+validation 2	Session 3+session4+Session 7+8
Present the work developed as a team in a relevant and convincing way.	Final validation	Session 14